

ALTAI, Mongolia

WMO: 442770

Lat: 46.3775N

Lon: 96.2378E

Elev: 2181

StdP: 77.72

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.8	-29.2	-35.5	0.2	-29.5	-32.9	0.2	-27.4	18.3	-13.5	16.0	-11.6	1.2	270	0.784

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	25.1	13.6	23.4	12.7	21.9	12.3	16.9	22.4	15.5	20.8	14.2	19.5	4.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.8	13.8	20.3	13.3	12.5	18.7	12.0	11.4	17.6	56.7	22.5	51.7	21.0	47.5	19.5	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	13.0	10.9	-34.3	27.7	2.7	1.4	-36.2	28.7	-37.8	29.5	-39.4	30.3	-41.4	31.3		
			-34.1	17.2	2.7	3.5	-36.1	19.7	-37.7	21.8	-39.2	23.8	-41.2	26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.4	-17.9	-14.8	-7.6	1.3	7.2	13.2	15.7	13.8	7.6	-0.5	-9.2	-15.0	(6)
(7)		DBStd	12.71	5.89	5.93	6.36	5.38	4.74	3.56	3.22	3.47	4.46	5.27	5.88	5.57	(7)
(8)		HDD10.0	4273	865	694	547	262	110	12	1	8	96	326	576	776	(8)
(9)		HDD18.3	6872	1123	927	805	511	344	157	93	145	323	584	826	1034	(9)
(10)		CDD10.0	462	0	0	0	2	24	108	179	126	23	0	0	0	(10)
(11)		CDD18.3	19	0	0	0	0	0	3	12	4	0	0	0	0	(11)
(12)		CDH23.3	150	0	0	0	0	1	27	94	29	0	0	0	0	(12)
(13)		CDH26.7	12	0	0	0	0	0	2	9	1	0	0	0	0	(13)
(14)	Wind	WSAvg	3.9	3.0	3.2	4.3	5.3	5.1	4.0	3.5	3.6	4.1	4.1	3.9	3.3	(14)
(15)	Precipitation	PrecAvg	170	1	2	5	10	14	29	41	38	16	8	3	2	(15)
(16)		PrecMax	307	6	11	16	30	49	104	117	101	98	34	10	7	(16)
(17)		PrecMin	95	0	0	0	0	0	2	3	2	0	0	0	0	(17)
(18)		PrecStd	54	1	2	5	6	11	19	28	25	16	7	3	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.9	1.3	9.4	17.2	22.2	26.3	28.0	25.9	21.2	14.1	7.2	0.1	(19)
(20)			MCWB	-3.8	-2.2	2.4	8.6	11.5	12.9	16.6	13.7	10.0	5.6	1.9	-3.5	(20)
(21)		2%	DB	-3.9	-1.5	6.6	14.1	19.9	23.7	25.8	23.9	18.8	11.5	4.4	-2.3	(21)
(22)			MCWB	-6.4	-4.5	0.4	5.4	10.0	12.4	14.5	12.5	9.2	4.4	0.0	-4.9	(22)
(23)		5%	DB	-6.0	-3.2	4.4	12.0	17.7	22.1	24.1	22.2	16.9	9.5	1.8	-4.3	(23)
(24)			MCWB	-8.2	-5.9	-1.0	4.1	8.3	12.1	13.5	12.3	8.5	3.2	-1.7	-6.5	(24)
(25)		10%	DB	-8.3	-5.2	2.1	9.9	15.7	20.5	22.4	20.6	15.0	7.7	-0.4	-6.4	(25)
(26)			MCWB	-9.9	-7.5	-2.3	3.0	7.2	11.1	12.9	11.8	7.6	2.2	-3.2	-8.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.4	-1.8	4.0	9.6	14.0	16.7	19.8	17.8	13.1	7.9	2.5	-2.8	(27)
(28)			MCDWB	-1.7	0.6	7.9	15.7	21.1	23.7	26.1	23.1	19.0	13.2	5.8	-0.6	(28)
(29)		2%	WB	-6.1	-4.2	1.1	6.9	11.5	15.1	17.7	16.0	11.3	5.5	0.4	-4.7	(29)
(30)			MCDWB	-4.2	-1.9	5.4	11.8	18.0	21.5	23.6	20.9	16.3	9.7	3.9	-2.5	(30)
(31)		5%	WB	-8.2	-5.8	-0.8	5.0	9.7	13.8	16.2	14.7	9.7	4.0	-1.4	-6.5	(31)
(32)			MCDWB	-6.3	-3.5	3.6	10.2	15.6	19.8	21.6	19.7	14.4	8.3	1.6	-4.4	(32)
(33)		10%	WB	-10.0	-7.5	-2.2	3.4	8.1	12.6	14.9	13.4	8.3	2.6	-3.1	-8.3	(33)
(34)			MCDWB	-8.2	-5.2	2.0	9.0	13.8	18.0	19.6	18.0	13.5	7.0	-0.6	-6.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.8	12.3	11.9	10.5	10.8	10.6	9.8	10.1	10.5	10.5	10.6	11.0	(35)
(36)			MCDBR	12.4	13.1	12.6	12.4	12.9	12.9	11.8	12.0	11.8	12.0	11.6	12.1	(36)
(37)			MCWBR	10.5	10.7	8.6	7.4	7.4	6.8	5.8	6.1	6.5	7.7	8.6	10.1	(37)
(38)		5% WB	MCDWB	12.4	12.7	12.2	11.8	12.2	11.4	10.6	10.9	10.9	11.6	11.4	11.8	(38)
(39)			MCWBR	10.9	10.7	9.0	8.4	8.4	7.6	6.9	7.1	7.4	8.4	9.0	10.1	(39)
(40)		Clear-Sky Solar Irradiance	taub	0.230	0.261	0.305	0.346	0.350	0.360	0.356	0.333	0.299	0.280	0.250	0.233	(40)
(41)	taud		2.369	2.287	2.204	2.173	2.225	2.278	2.331	2.378	2.432	2.402	2.385	2.355	(41)	
(42)	Ebn at Noon		898	922	919	907	913	903	902	913	920	889	857	848	(42)	
(43)	Edh at Noon		70	94	121	137	136	130	121	111	95	83	68	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.91	3.04	4.56	5.99	7.03	7.03	6.74	6.00	5.09	3.46	2.08	1.55	(44)	
(45)		RadStd	0.14	0.13	0.22	0.24	0.41	0.37	0.32	0.25	0.19	0.18	0.18	0.10	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days					
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	-3.28	N/A	-2.26	-2.41	N/A	N/A	N/A	N/A	(46)		
(47)	Regional (5 neighbors)	N/A	N/A	-2.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)		

Nomenclature: See separate page